## Sequences

**1** For each of the following sequences, find the indicated term:

**a** Third term in the sequence: $2, -4, 6, -8, 16, \ldots$

**b** Fourth term in the sequence: $3, 3.5, 4, 4.5, 5, 5.5, \ldots$

**c** Fifth term in the sequence: $5, 4, 3, 2, 1, 0, -1, \ldots$

**2** If $T_n$ describes the $n$th term of the following sequences , find the indicated term:

**a** $T_3$ in the sequence: $4, -5, 6, -7, 8, \ldots$

**b** $T_4$ in the sequence: $200, 20, 2, 0.2, 0.02, \ldots$

**c** $T_5$ in the sequence: $1, 2.5, 4, 5.5, 7, 8.5, \ldots$

**d** $T_3 + T_5$ in the sequence: $6, -8, 9, -10, 11, \ldots$

**e** $2T_2 - T_4$ in the sequence: $9, 12, 15, 18, 21, \ldots$

**f** $-4(T_3 + T_4)$ in the sequence: $1, 4, 5, 9, 14, 23, \ldots$

**3** In the sequence $5, -5, 7, -7, 9, -9, \ldots$, find $n$ if $T_n = 7$.

**4** For each of the following sequences:

**i** Describe the pattern in words.

**ii** Find the next number in the sequence.

**a** $-1, 1, 3, 5, 7, \ldots$

**b** $64, 32, 16, 8, 4, 2, \ldots$

**c** $2, -4, 6, -8, 10, -12, \ldots$

## Explicit rules

**5** Determine whether the following are explicit relations:

**a** $b_n = b_{n-1}b_{n-2}$

**b** $d_n = n^2 + 3n + 8$

**c** $a_n = 8a_{n-1} + a_1$

**d** $c_n = 3n^2$

**6** For each of the following explicit rules, state the first $5$ terms of the sequences in order starting with $n = 1$:

**a** $b_n = 5n - 2$

**b** $s_n = n^2 + 6$

**c** $t_n = 2n^2 + n - 3$

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**7** For the following sequencesn state the explicit rule that describes $T_n$ in terms of $n$.

**a** A sequence starts with a first term of $1300$ and each subsequent term increases by $2.5\%$ of the previous term.

**b** A sequence starts with a first term of $44$ and each term is $77$ more than the previous term.

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